

Improving Evidence Use and Semantic Verification in Multi-Agent Business Analytics

A Failure-Guided Controlled Study

Mikhail Dushkin

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上海交通大学
SHANGHAI JIAO TONG UNIVERSITY

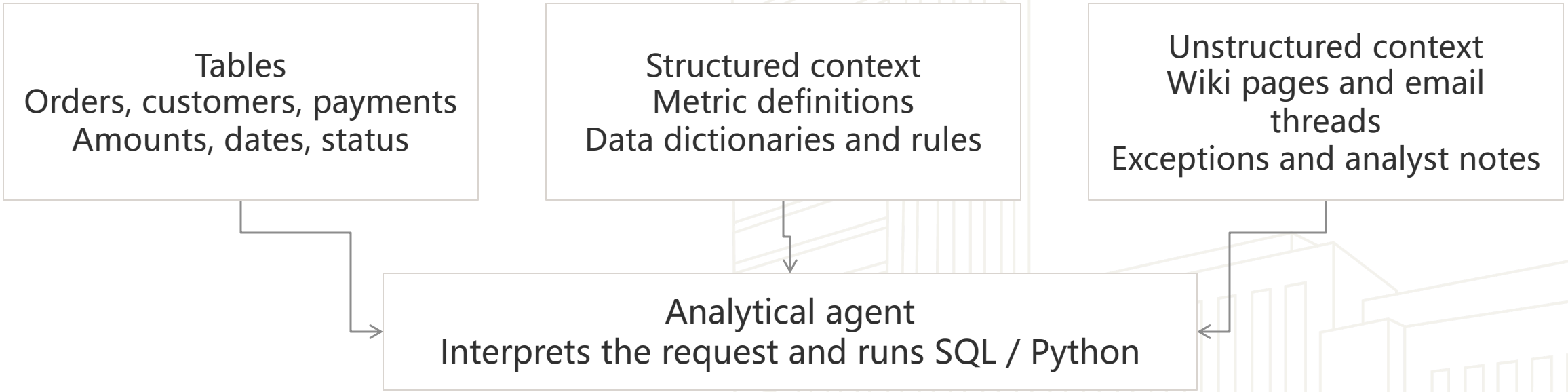
安泰经管学院
ANTAI COLLEGE
Economics · Management

Master of International Business
Supervisor: Prof. Liu Jialu

Background

Company data and business context

A manager asks: “What were our May sales?”



The agent needs the applicable business meaning, as well as the records.

Background

Business rules change the calculation

Company rule: completed orders count in the month of completion.

Order	Amount	Order date	Completion date	Status
A	100	20 May	22 May	Completed
B	80	21 May	—	Cancelled
C	50	29 May	2 June	Completed

All May orders
230 → Completed orders
150 → Completed in May
100

Illustrative company rule and data. All three calculations can execute successfully.

Background

Two ways the wrong answer survives

1 The analyst repeats a past calculation under the wrong conditions

Past procedure Sum by order date	→	Analyst reuses it Current rule is ignored	→	Reported May sales 230 instead of 100

Evidence-checked experience reuse

Check whether the old procedure applies to the current sources.

2 The reviewer approves the number after checking only the computation

Source-grounded counterexample review

Test the disputed business interpretation with a small distinguishing example.

Research Question

Reliability and cost-effectiveness

How do **evidence-checked experience reuse** and **source-grounded counterexample review**, **separately and together**, affect the reliability and cost-effectiveness of a hierarchical multi-agent system for business analytics?

Compare the targeted errors and final answers.
Test the hierarchy against strong single-agent controls.

MAST: a framework for describing failures

Cemri et al. (2025), “Why Do Multi-Agent LLM Systems Fail?”

Multi-Agent System Failure Taxonomy: 14 failure modes in three groups

System design / Inter-agent misalignment / Task verification

Evidence and experience use

FM-2.4: a finding is not passed on.
FM-2.5: received evidence is ignored.
Experience misuse may require finer coding.

Semantic verification

FM-3.2: a required check is omitted.
FM-3.3: an attempted check is judged wrongly.

Classify observed actions; use controlled comparisons to test the repairs.

Cemri et al. · MAST v3

Literature Review

DABstep: a common test for analytical agents

Data Agent Benchmark for Multi-step Reasoning

Public dataset and competitive leaderboard by Adyen and Hugging Face

Inputs	→	Agent task	→	Evaluation
Payment tables and supporting documents		Find rules, combine data and calculate an answer		Submit final answers for accuracy scoring

Example task type: identify applicable fee rules and calculate merchant fees.

450 evaluation questions · Answers include numbers, short text and lists

DABstep paper v1 · Hugging Face dataset · Public leaderboard

Literature Review

Existing systems and mechanisms

Reference	Relevant approach	Role in our study
DataPilot	Scoped assets and procedures	Experience-use reference
DS-STAR	Planning and iterative review	Review-workflow reference
NVIDIA	Prepared tools; one inference agent	Strong single-agent reference
FinHelm	Memory, delegation and verification	Public engine to adapt

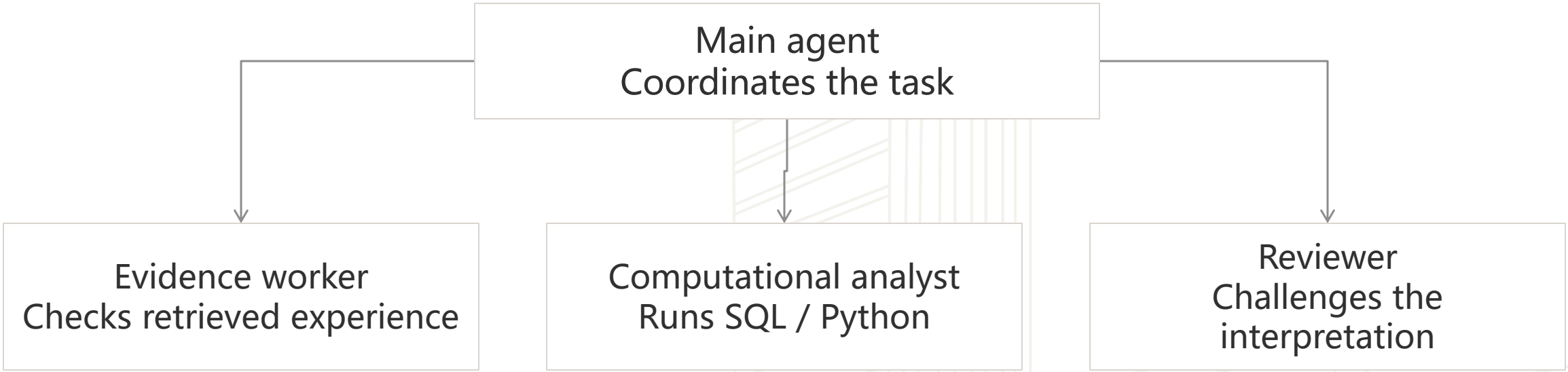
SQLDriller / TS-SQL: executable tests · QueryProof: business-rule gates

DataPilot, DS-STAR and NVIDIA were high-performing in our 14 October audit.
FinHelm is an implementation starting point; reproducing their scores is not claimed.

DataPilot · DS-STAR v4 · NVIDIA Data Explorer · FinHelm · pinned revision · SQLDriller · TS-SQL · QueryProof v1

Research Plan & Method

A fixed hierarchy adapted from FinHelm

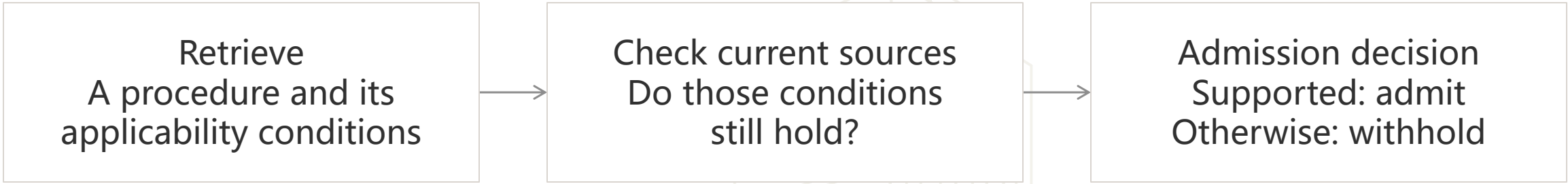


Multi-agent system (MAS): delegated work in separate child contexts, with observable results transferred back to the main agent.

FinHelm · pinned revision

Research Plan & Method

Evidence-checked experience reuse



Past procedure: sum orders by order date.

Current rule: completed orders, using completion date.

Decision: withhold this procedure and analyse the current sources.

Measure inappropriate reuse and the retention of useful experience.

DataPilot · ReasoningBank v2

Research Plan & Method

Source-grounded counterexample review

The reviewer reads the question and sources before seeing the candidate.

Small test: one order worth 50, placed 29 May and completed 2 June

Interpretation	Contribution to May sales
Use order date	50
Use completion date	0

The documented completion-date rule requires 0 for May.

Test validity and expected answers remain fallible.

SQLDriller · TS-SQL · QueryProof v1

Research Plan & Method

Controlled experiment and evaluation

Condition	Experience check	Test-based review
Baseline	Off	Off
Evidence-checked experience reuse	On	Off
Source-grounded counterexample review	Off	On
Both	On	On

Strong single-agent controls: same model, sources, tools, assets and aggregate caps.

DABstep final-answer quality + trace diagnosis + identical reviewer candidates
Estimate separate and joint effects; a combined win alone is not interaction.

Expected Results & Innovation Points

Research contribution and practical use

Research contribution

Explain where business knowledge is lost or misapplied.

Test each mechanism and their combination in one reproducible system.

Refine MAST only where recurring trace evidence supports it.

Practical use

Help teams choose which checks to enable.

Show when preparing and reusing knowledge is worthwhile.

Clarify when a simpler single agent is sufficient.

Expected outputs: failure evidence, an implementation and quality–cost comparisons.

Expected Results & Innovation Points

Cost-effectiveness and business decisions

Operating cost

Model and tool spending per task
Latency, failed runs and retries
Cost per correct answer, with coverage

Preparation and adaptation

Hours to prepare knowledge and memory
Effort to adapt to a new task setting
Amortisation over future query volume

Does additional reliability justify the extra work?

Compare quality at the same budget and cost at comparable quality.
State preparation and query-volume assumptions explicitly.

Challenges & Limitations

What may limit the benefits?

1. Analyst and reviewer can share the same wrong interpretation.
2. Generated tests or their expected answers can be invalid.
3. Useful experience may be rejected; checks may cost more than they save.
4. Too few target failures can limit what the experiment establishes.
5. One hierarchy and benchmark cannot represent every business setting.

Expected Research Progress

Prepared work and next steps

Prepared: literature synthesis, problem map, research design and proposal.

Period	Planned work
Late October	Minimal harness; agree model and budget
November	Implement, validate, pilot and freeze the protocol
December	Main comparisons and midterm report
January	Failure, uncertainty and cost analysis
February–April	Write, revise and submit the thesis
May	Final defense

Which checks are worth using in business analytics?